

Fractions & Decimals

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS A fraction and a decimal can name the **same value two ways**. Common ones: $\frac{1}{4} = 0.25$, $\frac{1}{2} = 0.5$, $\frac{3}{4} = 0.75$. Tenths and hundredths connect them: $\frac{5}{10} = 0.5$ and $\frac{25}{100} = 0.25$. A **number line** and a **10×10 grid** help you see the match: shade 50 of 100 squares to show $\frac{5}{10} = 0.5$.

Round 1 — Retrieve It

1. Write $\frac{1}{2}$ as a decimal.

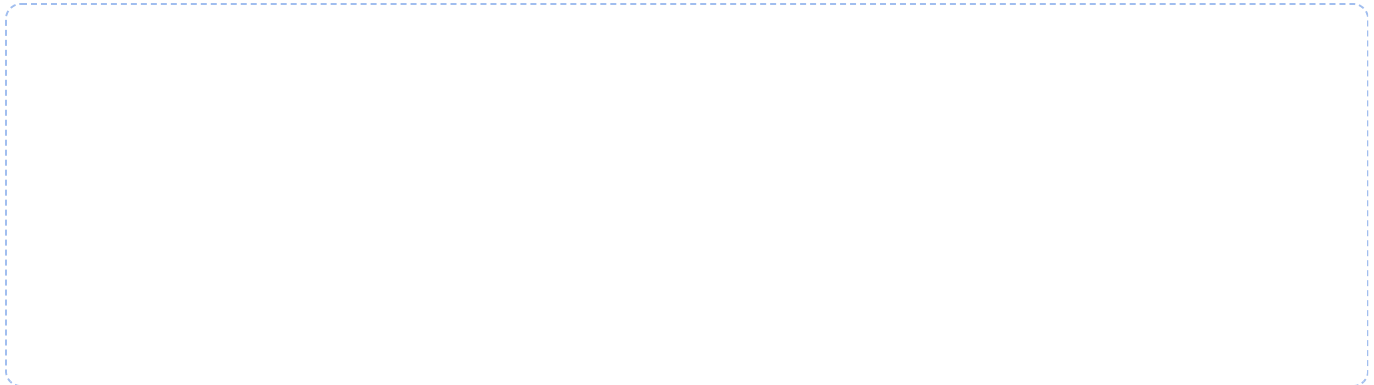
2. Write 0.75 as a fraction.

3. $\frac{5}{10}$ equals what decimal?

4. Name two tools that help you see a fraction and decimal are equal.

Round 2 — Sketch It

Draw a 10×10 grid. Shade squares to show $\frac{5}{10}$. Write the fraction and the decimal it equals below the grid.



Fractions & Decimals — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. Write $\frac{1}{4}$ as a decimal. What part of a whole is that?

2. Write 0.5 as a fraction two ways (tenths and simplest form).

3. On a number line from 0 to 1, where does $\frac{3}{4}$ fall? What decimal is that?

Round 4 — Reason from Evidence

Tom says $\frac{1}{4}$ is the same as 0.4 because he sees a 4 in each. Is he correct? Use a grid or the value of the places to explain the true decimal for $\frac{1}{4}$.

Fractions & Decimals — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how a fraction and a decimal can name the same value. Use the words tenths, hundredths, and equal.

C**Connect**

Connect fractions and decimals to money. Show how $\frac{1}{4}$ of a dollar equals \$0.25.

E**Extend**

Pick a fraction and write it as a decimal, then show it on a 10×10 grid or a number line. Explain the match.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Fractions & Decimals

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. 0.5.
2. $\frac{3}{4}$ (also $\frac{75}{100}$).
3. 0.5.
4. A number line and a 10×10 grid.

Sketch It — look for

A 10×10 grid with 50 of the 100 squares shaded. Labeled $\frac{5}{10} = 0.5$ (also $\frac{1}{2}$).

Apply It

1. 0.25. It is one fourth (25 hundredths) of a whole.
2. $\frac{5}{10}$ and $\frac{1}{2}$.
3. $\frac{3}{4}$ falls three fourths of the way from 0 to 1, at 0.75.

Reason from Evidence — look for

No. $\frac{1}{4} = \frac{25}{100} = 0.25$, not 0.4. On a grid, $\frac{1}{4}$ shades 25 squares; 0.4 would shade 40. The digits alone do not tell the value.

ACE — Extend look-for

A correct fraction-decimal pair (such as $\frac{1}{2} = 0.5$ or $\frac{3}{4} = 0.75$) shown on a grid or number line, with an explanation that both name the same amount.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.